

Homework 4

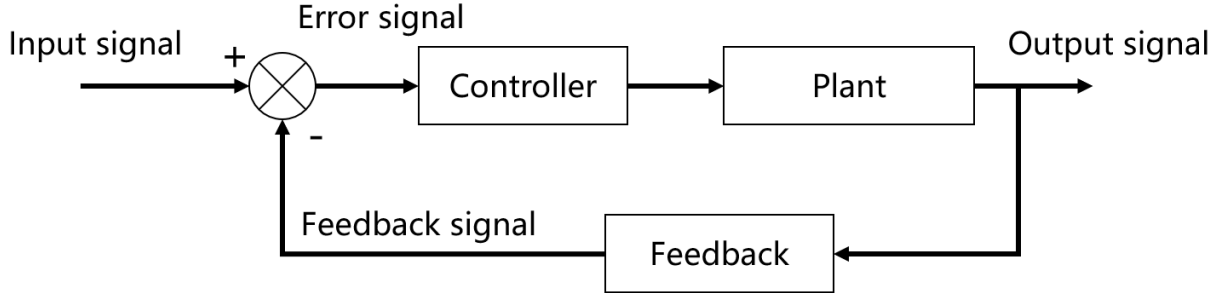
Jizhou Yan ^{TA}

Instructions before the assignment 作业前须知:

- This assignment is for the **fifth week's homework** of the System Dynamics and Control course (hereinafter referred to as SDC). The submission deadline is **before 11:59 PM on October 17, 2024**. Please ensure timely submission, as late submissions will result in point deductions. 本次作业为 SDC 第五周课程作业, 提交时间为 **2024/10/17 23:59** 前。
 - If submitted within 12 hours after the deadline, no points will be deducted (**this is allowed only twice, so focus on completing it properly rather than rushing to meet the deadline if you're unable to finish on time**);
 - If submitted within three days after the deadline, **20% of the total score of this assignment will be deducted**;
 - If submitted within two week after the deadline, **30% of the total score of this assignment will be deducted**;
 - Any other late submissions will result in a **40% deduction**;
- For late submissions, please contact the teaching assistant(TA), **Yan Jizhou**, via email at **yanjizhou312@163.com** or WeChat **y3139838452**. Make sure to use the filename format as "SDC-Assignment-x-Resubmission-Name-Student ID." 作业补交时通过邮箱或微信与助教闫济洲联系, 邮件标题及文件名备注为 "SDC-第 x 次作业补交-姓名-学号".
- The assignment is graded out of 10 points. It consists of basic course questions (10 points) and extension questions (0 to 5 points). **The final assignment score will be the smaller one between 10 and the actual score.** Each assignment earns all points by completing only the basic assignment. 每次作业仅完成基础作业即可获得所有分数, 鼓励同学们完成拓展课程题目。
- Students are encouraged to attempt the extension questions. Any extra points earned will be proportionally added to the final course grade for possible grade improvement.
- **Academic misconduct such as plagiarism is strictly prohibited in course work. Plagiarism and other academic misconduct will be dealt with in accordance with the relevant regulations of the University.** If you have discussed the assignment with classmates or received help, please note at the beginning of the assignment: "This assignment was completed after discussion with XXX (classmate's name), and the thought process was referenced from XXX (classmate's name)." **Every contribution deserves recognition. Unacknowledged similarities between assignments will be treated as plagiarism.** 课程作业中严禁抄袭等学术不端行为。如作业完成过程中经历了同学讨论/同学帮助的情况, 请在作业最前面注明 "本次作业经过与 XXX (同学姓名) 讨论, 参考了 XXX (同学姓名) 的部分思路"。每位同学的帮助都值得被记录。不被注明的作业雷同将被认为是抄袭行为。

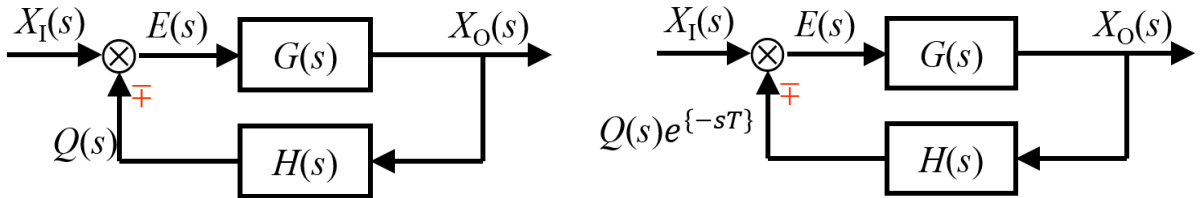
1. Problem 1

In this week's lesson, we learned how to represent the process of control and modeling using block diagrams. We successfully represented both our control process and modeling process with block diagrams. This reminded us of the control systems from our first class.



Here, we need to consider a question: Are the block diagrams used in the control process and the modeling process exactly the same and consistent?

In practical situations, the feedback process in the control block diagram is obtained from measurements using methods such as sensors. This inevitably means that the feedback applied to the input signal actually comes from the output signal of the previous frame. Although we generally ignore this very small time delay, it still exists. After performing a Laplace transform, this will introduce a factor of e^{-sT} into the feedback, where T is the sampling time. When T is very close to 0, the effect of this factor can be neglected.



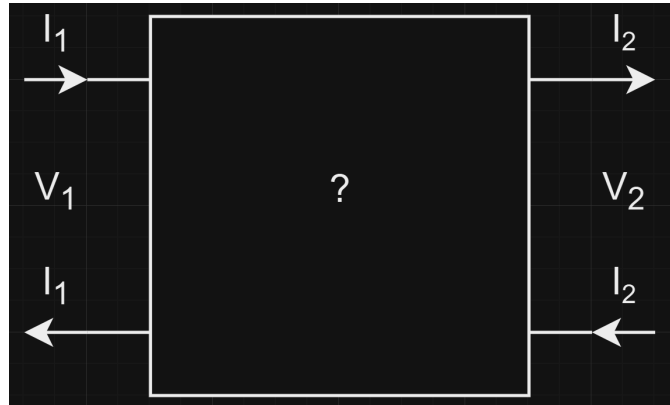
In the modeling process, feedback loops mostly arise from the internal physical relationships of the system. For example, in a DC motor, the relationship between current and torque is given by $T = K_T i$, and the relationship between back electromotive force and angular velocity is $e = K_b \omega$. This represents the conversion from electrical energy to mechanical energy. Due to the law of conservation of energy, the processes of converting current to torque and angular velocity to electromotive force occur simultaneously, which defines the loop of the DC motor.

This reminds us that the processes of generating torque from current and producing electromotive force from angular velocity are essentially the same, merely described by different physical quantities. This further suggests that we might construct a transfer function for the motor as a whole, rather than breaking it down into the current-torque and angular velocity-electromotive force subsystems. This approach could better illustrate the energy conversion process and may enhance our understanding of system modeling.

Next, we will explore the following question: **Can we write the transfer function in a more complex form that directly represents the energy conversion and transfer process between the input and output of a system?**

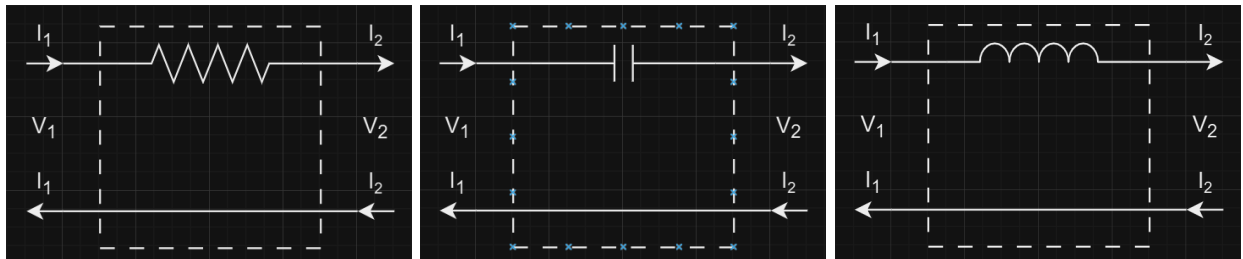
Fortunately, we are not starting from scratch. In electronic systems, a two-port network is represented as a network with two pairs of terminals connected to external circuits. The two-port network model is used in mathematical circuit analysis techniques to isolate portions of larger circuits. A two-port network is regarded

as a "black box" with its properties specified by a matrix of numbers. This allows the response of the network to signals applied to the ports to be calculated easily, without solving for all the internal voltages and currents in the network. It also allows similar circuits or devices to be compared easily.



There are many parameters that can be used to describe a two-port network, with several commonly used sets including impedance parameters Z , admittance parameters Y , hybrid parameters h and g , and transmission parameters T . Here, we will mainly focus on the transmission parameters T . For those interested in learning more about other parameters, I recommend reading the renowned white paper "Fundamentals of Electronic Circuits and Systems" from the Department of Electronics, or checking out wikipedia.org

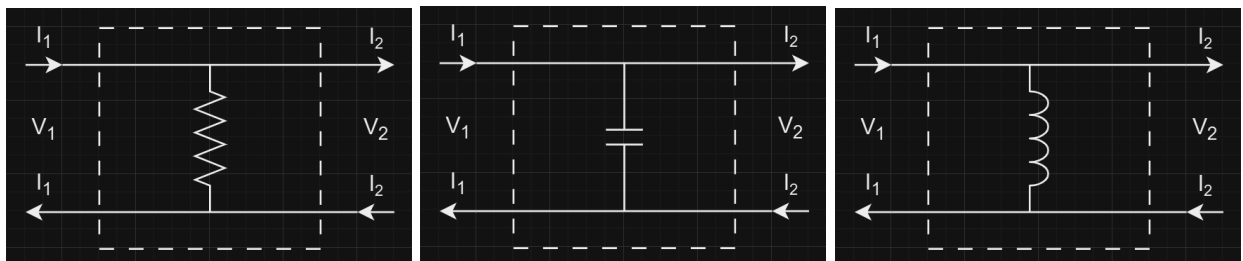
In the circuit, the transmission parameters T are described as the relationship between the voltage and current at the two ports (Miss Elysia passed by and helped you perform the Laplace transform.).



$$\begin{bmatrix} U_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 & -R \\ \cdot & 1 \end{bmatrix} \begin{bmatrix} U_1 \\ I_1 \end{bmatrix}$$

$$\begin{bmatrix} U_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{1}{sC} \\ \cdot & 1 \end{bmatrix} \begin{bmatrix} U_1 \\ I_1 \end{bmatrix}$$

$$\begin{bmatrix} U_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 & sL \\ \cdot & 1 \end{bmatrix} \begin{bmatrix} U_1 \\ I_1 \end{bmatrix}$$

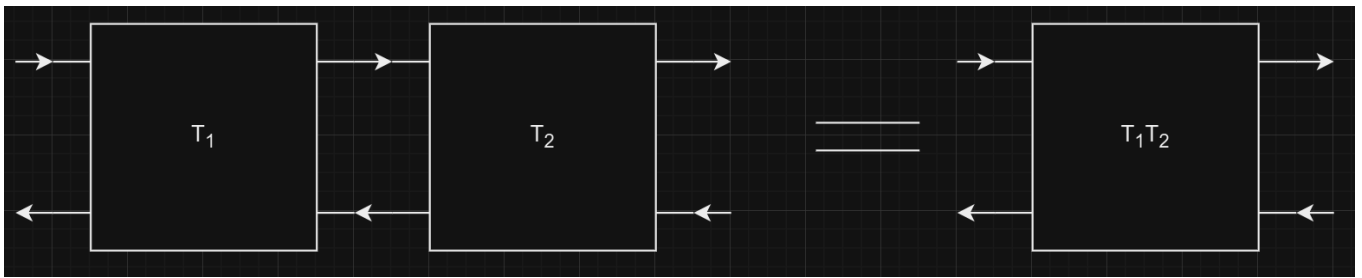


$$\begin{bmatrix} U_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 & \cdot \\ -\frac{1}{R} & 1 \end{bmatrix} \begin{bmatrix} U_1 \\ I_1 \end{bmatrix}$$

$$\begin{bmatrix} U_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 & \cdot \\ -sC & 1 \end{bmatrix} \begin{bmatrix} U_1 \\ I_1 \end{bmatrix}$$

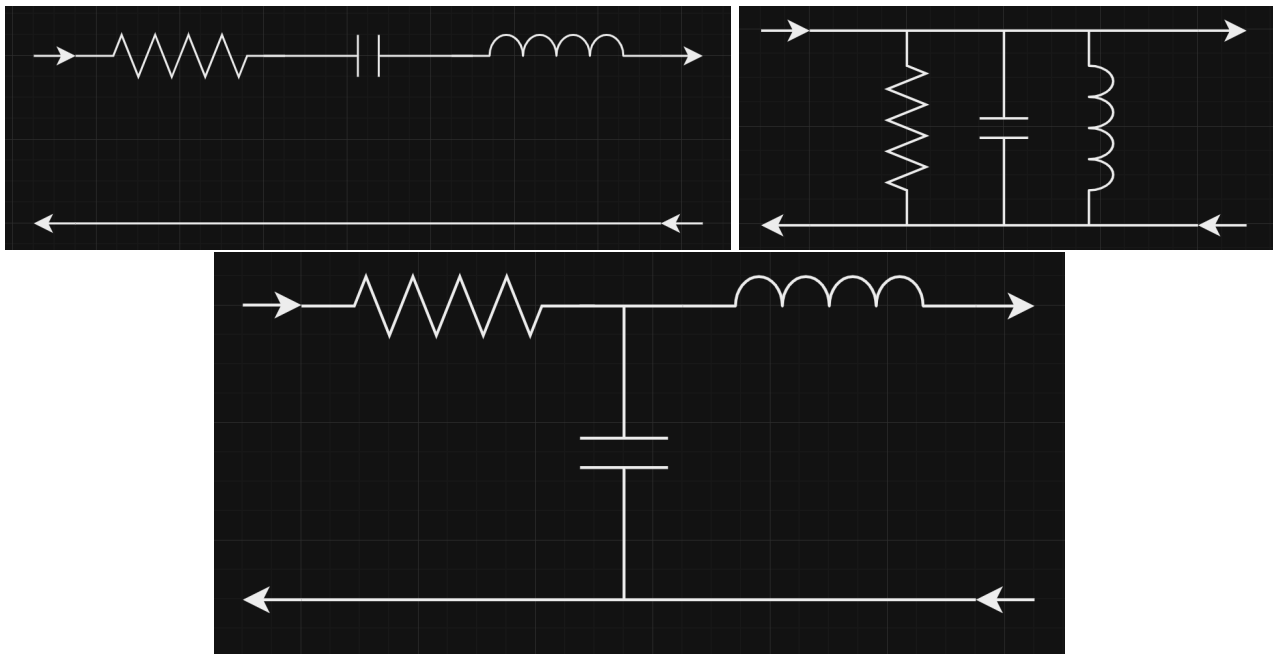
$$\begin{bmatrix} U_2 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 & \cdot \\ -\frac{1}{sL} & 1 \end{bmatrix} \begin{bmatrix} U_1 \\ I_1 \end{bmatrix}$$

The usage of the transmission matrix is consistent with that of the transfer function; you connect the endpoints using multiplication.



1.1 a

Provide the corresponding transmission matrix T for the following two-port circuit network:



1.2 b

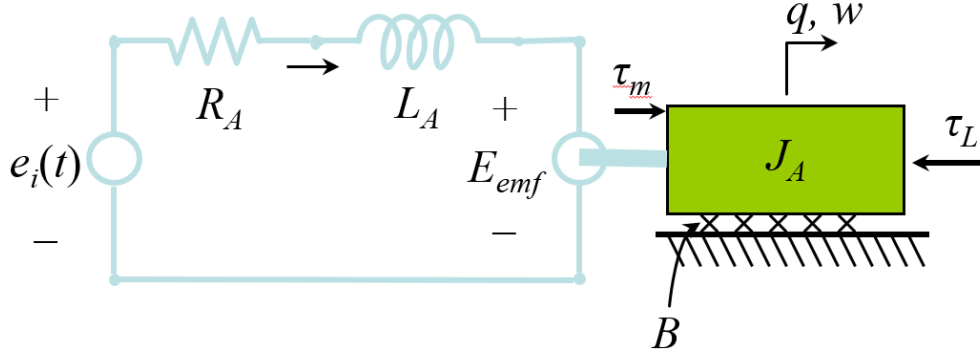
Next, we consider the transmission process from the circuit system through the motor to the mechanical system. We define the output port quantities of the motor as $\Phi = \begin{bmatrix} \Omega \\ T \end{bmatrix}$.

Based on the input-output relationships of the motor, $T = K_T i$ and $u = K_b \omega$, we need to construct the transmission matrix T such that:

$$\Phi = T \cdot \begin{bmatrix} u \\ i \end{bmatrix}$$

1.3 c

Let's take a look at the original question from the course.



"Please provide the transmission matrix T_1 from the power source $\Phi_i = \begin{bmatrix} E_i \\ I \end{bmatrix}$ to the motor output $\Phi_o = \begin{bmatrix} \Omega \\ T_m \end{bmatrix}$, as well as the transmission matrix T_2 from the external load $\Phi_L = \begin{bmatrix} \Omega \\ T_L \end{bmatrix}$ to the motor output $\Phi_o = \begin{bmatrix} \Omega \\ T_m \end{bmatrix}$. We can derive an equation, can't we?"

$$T_1 \Phi_i = \Phi_o = T_2 \Phi_L$$

(Expansion) Is it possible to derive an equation that includes only E_i , T_m , Ω , and the device parameters? How does this equation compare to the transfer function obtained during the course?"

2. Problem 2

The ECP plant, shown in Figure 1 is designed to emulate a broad range of typical servo control applications. The Model 220 apparatus consists of a drive motor (servo actuator) which is coupled via a timing belt to a drive disk with variable inertia. Another timing belt connects the drive disk to the speed reduction (SR) assembly while a third belt completes the drive train to the load disk. The load and drive disks have variable inertia which may be adjusted by moving (or removing) brass weights. Speed reduction is adjusted by interchangeable belt pulleys in the SR assembly. Backlash may be introduced through a mechanism incorporated in the SR assembly, and flexibility may be introduced by an elastic belt between the SR assembly and the drive disk. The drive disk moves one-for-one with the drive motor so that its inertia may be thought of as being collocated with the motor. The load inertia however will rotate at a different speed than the drive motor due to the speed reduction. Also, drive flexibility and/or backlash may exist between the load and the drive motor and hence its inertia is considered to be non-collocated with the motors. All rotating shafts of the mechanism are supported by precision ball bearings. Needle bearings in the SR assembly provide low friction backlash motion (when backlash is desired).

A disturbance motor connects to the load disk via a 4:1 speed reduction and is used to emulate viscous friction and disturbances at the plant output. A brake below the load disk may be used to introduce Coulomb friction. Thus friction, disturbances, backlash, and flexibility may all be introduced in a controlled manner. These effects represent non-ideal conditions that are present to some degree in virtually all physically realizable electromechanical systems.

High-resolution incremental encoders couple directly to the drive (1) and the load (2) disks providing position (and derived rate) feedback. The drive and disturbance motors are electrically driven by servo amplifiers and power supplies in the Controller Box. The encoders are routed through the Controller box to interface directly with the DSP board via a gate array that converts their pulse signals to numerical values.

The schematic of the plant with rigid drive belts is shown in Fig.2, in which TD is the electromagnetic

torque produced by the drive motor (the inertia and the damping effect of the motor shaft have been lumped into the total drive motor inertia J_d and c_l shown in Fig. 2), and T_L is the lumped disturbance torque generated by the load motor including various un-modeled Coulomb frictions. Assume that the motor armature coil has a resistance of R_A and a negligible inductance (i.e., $L_A=0$). Based on the schematic, obtain the system dynamic model as follows

2.1 a

Identify reference points and positive directions and draw FBD for each mechanical element shown in Fig.2 and the diagram for electrical circuit. Assuming an ideal pulley (i.e., $J_p=0$) for simplicity, write down all the elemental equations and interconnecting equations. Check to make sure that the number of equations is the same as the number of unknown variables.

2.2 b

Obtain the transfer functions of individual subsystems to draw the block diagram of the overall system showing the functionality of each element (the electrical, the EM coupling, the drive disk, the ideal pulley, and the load disk). Output of the overall system is assumed as θ_1 .

2.3 c

Obtain the transfer functions of the overall system from the block diagram in (2), and write down the input/output model of the overall system in time-domain.

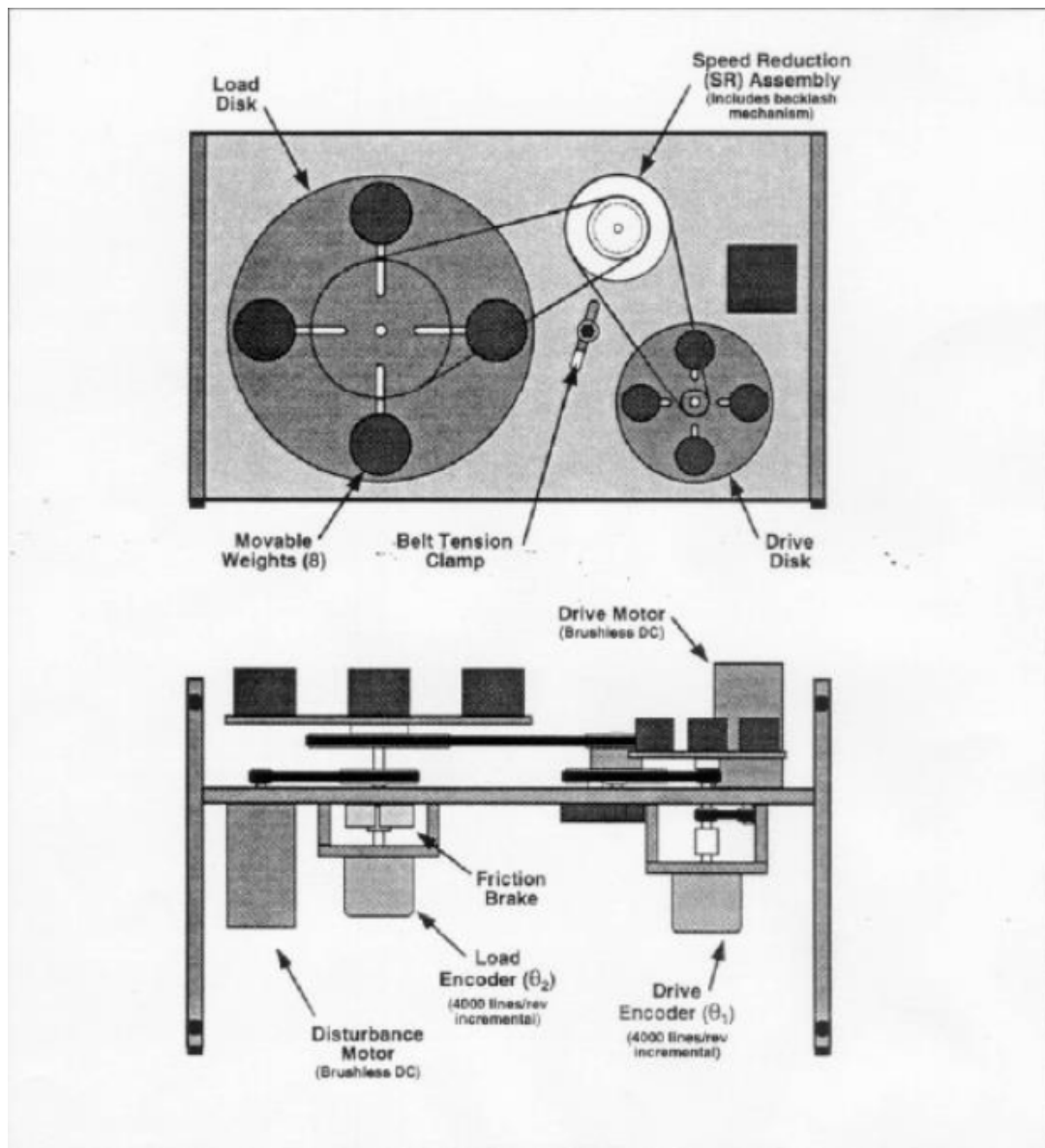


Figure 1. mulator Apparatus

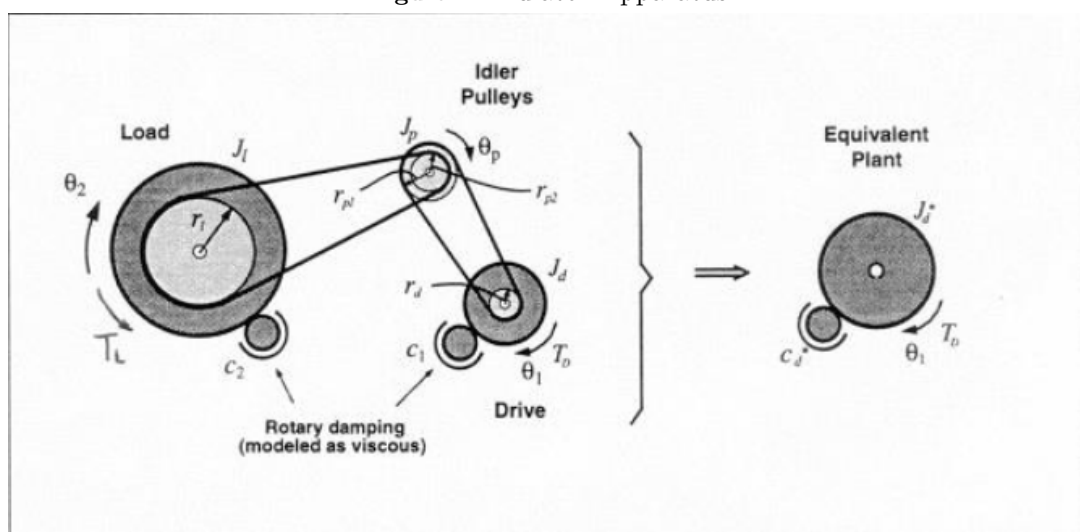


Figure 2. Schematic of ECP Plant with Rigid Belts